1. **Rice (Cammeo and Osmancik)**

Among the certified rice grown in TURKEY, the Osmancik species, which has a large planting area since 1997 and the Cammeo species grown since 2014 have been selected for the study. When looking at the general characteristics of Osmancik species, they have a wide, long, glassy and dull appearance. When looking at the general characteristics of the Cammeo species, they have wide and long, glassy and dull in appearance. A total of 3810 rice grain's images were taken for the two species, processed and feature inferences were made. 7 morphological features were obtained for each grain of rice.

2. **Heart Disease**

This database contains 76 attributes, but all published experiments refer to using a subset of 14 of them. In particular, the Cleveland database is the only one that has been used by ML researchers to date. The "goal" field refers to the presence of heart disease in the patient. It is binary valued. Experiments with the Cleveland database have concentrated on simply attempting to distinguish presence (value 1) from absence (value 0).

3. **Dry Bean Dataset**

Two different types of dry beans were used in this research, taking into account the features such as form, shape, type, and structure by the market situation. A computer vision system was developed to distinguish two different registered varieties of dry beans with similar features in order to obtain uniform seed classification. Bean images obtained by computer vision system were subjected to segmentation and feature extraction stages, and a total of 16 features; 12 dimensions and 4 shape forms, were obtained from the grains.

4. **Wine**

These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from two different cultivars. The analysis determined the quantities of 13 constituents found in each of the two types of wines. In a classification context, this is a well-posed problem with "well behaved" class structures.

5. **Breast Cancer Wisconsin (Diagnostic)**

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image. Relevant features were selected using an exhaustive search in the space of 1-4 features and 1-3 separating planes to perform a classification based on decision trees

6. **Car Evaluation**

Car Evaluation Database was derived from a simple hierarchical decision model originally developed for the demonstration of DEX, M. Bohanec, V. Rajkovic: Expert system for decision making. Sistemica

1(1), pp. 145-157, 1990.). The model evaluates cars according to the several concept structure. Because of known underlying concept structure, this database may be particularly useful for testing constructive induction and structure discovery methods.

7. **Bank Marketing**

The data is related with direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to access if the product (bank term deposit) would be ('yes') or not ('no') subscribed. The classification goal is to predict if the client will subscribe (yes/no) a term deposit (variable y)

8. **Predict students' dropout and academic success**

The dataset was created in a project that aims to contribute to the reduction of academic dropout and failure in higher education, by using machine-learning techniques to identify students at risk at an early stage of their academic path, so that strategies to support them can be put into place. The dataset includes information known at the time of student enrollment – academic path, demographics, and social-economic factors. The problem is formulated as a two-category classification task (dropout and graduate) at the end of the normal duration of the course.

9. **Diabetes detection**

The diagnostic, binary-valued variable investigated is whether the patient shows signs of diabetes according to World Health Organization criteria (i.e., if the 2 hour post-load plasma glucose was at least 200 mg/dl at any survey examination or if found during routine medical care). The population lives near Phoenix, Arizona, USA.

10. **Drinking dataset**

This dataset is collected from National Health Insurance Service in Korea. All personal information and sensitive data were excluded. The purpose of this dataset is the classification of drinker based on analysis of body signal.